Physics 7B

UC Berkeley

Spring 2026

7B MidTerm 1 - Solutions

Topics

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Midterm 1 Solutions

Problem 1

1. The change in volume for any substance is given by the formula $\Delta V = V_{initial} \cdot \beta \cdot \Delta T$:

$$V_{metal} = V_0(1 + 3\alpha_m \Delta T) \quad (1)$$
$$V_{water} = V_0(1 + \beta_w \Delta T) \quad (2)$$
$$V_{container} = (2V_0)(1 + 3\alpha_c \Delta T) \quad (3)$$

Since the container remains exactly full without spilling, we have

$$V_{container} = V_{metal} + V_{water}$$

Substituting the expressions from the previous step:

$$(2V_0)(1 + 3\alpha_c \Delta T) = V_0(1 + 3\alpha_m \Delta T) + V_0(1 + \beta_w \Delta T)$$

Finally

$$6\alpha_c = 3\alpha_m + \beta_w$$

2. Since the container expansion is negligible, the equation simplifies to:

$$\beta_w = -3\alpha_m$$

Since the metal behaves regularly, α_m is a positive constant ($3\alpha_m > 0$). So β_w must be negative ($\beta_w < 0$).

Therefore, the experiment must have been conducted in the temperature range:

$$0°\mathrm{C} < T < 4°\mathrm{C}$$

. In this specific range, water undergoes anomalous expansion (it contracts as it warms). This contraction ($\Delta V_w < 0$) exactly offsets the regular thermal expansion of the metal block ($\Delta V_m > 0$), allowing the total volume of the contents to remain constant and the container to stay full to the top without spilling.

3. To determine the rate of heat flow $\frac{\delta Q}{dt}$, we apply Fourier's Law for a material where the cross-sectional area A varies as a function of x.

$$\frac{\delta Q}{dt} = -kA(x)\frac{dT}{dx}$$

The truncated pyramid has square faces. The side length $s(x)$ of the square cross-section increases linearly from $s(0) = a$ to $s(L) = 2a$:

$$s(x) = a(1 + \frac{x}{L})$$

The cross-sectional area $A(x)$ is therefore:

$$A(x) = a^2\left(1 + \frac{x}{L}\right)^2$$

In a steady state, the heat flow rate is constant throughout the material ($\frac{\delta Q}{dt} = C$). So :

$$-ka^2\left(1 + \frac{x}{L}\right)^2\frac{dT}{dx} = C$$

$$\frac{dT}{dx} = \frac{C}{-ka^2\left(1+\frac{x}{L}\right)^2}$$

We integrate both sides from the left face ($x = 0, T = T_1$) to the right face ($x = L, T = T_2$):

$$T_2 - T_1 = \frac{C}{-ka^2}\int_0^L \frac{1}{\left(1+\frac{x}{L}\right)^2}dx$$

$$T_2 - T_1 = \frac{C}{-ka^2}(\frac{-L}{2} + L)$$

$$\frac{\delta Q}{dt} = C = -\frac{2ka^2}{L}(T_2 - T_1)$$

Since the problem specifies that $T_2 > T_1$, $\frac{\delta Q}{dt}$ results in a negative value. It indicates that the heat flows in the **negative x-direction**. This is physically consistent with the second law of thermodynamics, which dictates that heat must flow from the higher temperature (T_2 at the right) to the lower temperature (T_1 at the left).

4.

$$\frac{\delta Q}{dt} = -kA(x)\frac{dT}{dx} = C$$

$$T(x) - T_1 = \frac{C}{-ka^2}\int_0^x \frac{1}{\left(1+\frac{x}{L}\right)^2}dx$$

$$T(x) - T_1 = \frac{2(T_2 - T_1)}{L}(\frac{-L}{1+\frac{x}{L}} + L)$$

$$T(x) - T_1 = \frac{2(T_2 - T_1)}{L}(\frac{x}{1+\frac{x}{L}})$$

$$T(x) = 2(T_2 - T_1)(\frac{\frac{x}{L}}{1+\frac{x}{L}}) + T_1$$

5. In a steady state, the heat flow rate $\frac{\delta Q}{dt}$ must be constant at every cross-section. Since the area $A(x)$ increases as x increases, the temperature gradient $\frac{dT}{dx}$ must **decrease** in magnitude to keep the product $A(x)\frac{dT}{dx}$ constant. A constant heat flow through a changing area results in a curved (non-linear) temperature distribution.

Problem 2

1. Since the mixture consists of both phases at equilibrium, the final temperature must be the fusion temperature

$$T_{final} = T_F$$

2. Heat absorbed by the solid

$$Q_{absorbed} = m_S C_S(T_F - T_S)$$

$$Q_{absorbed} = m_S(2\frac{L_F}{T_F})(T_F - \frac{T_F}{2})$$

$$Q_{absorbed} = m_S L_F$$

The resulting value for $Q_{absorbed}$ is **positive**. In thermodynamic convention, a positive sign indicates that heat energy is being transferred **into** the system. This confirms that the solid phase is absorbing thermal energy from the warmer liquid to increase its internal temperature.

3. Heat released by the liquid

$$Q_{released} = mC_L(T_L - T_F) + \frac{3}{4}mL_F$$
$$Q_{released} = mC_L(T_L - T_F) + \frac{3}{4}mL_F$$
$$Q_{released} = 2mL_F$$

The value for $Q_{released}$ is **positive**. By definition, when using the term "released," a positive sign indicates the quantity of energy exiting the liquid. This thermal energy is transferred to the cooler solid phase until the system reaches equilibrium.

4. Because the box is insulated, the heat released by the liquid is equal to the heat absorbed by the solid.

$$Q_{absorbed} = Q_{released}$$

Therefore,

$$m_s = 2m$$

5. The system's entropy change is the sum of the changes in the solid phase (ΔS_S) and the liquid phase (ΔS_L):

For the solid:

$$\Delta S_S = \int \frac{\delta Q}{T} = \int_{T_S}^{T_F} m_s C_s \frac{dT}{T})$$
$$\Delta S_S = 2m(2\frac{L_F}{T_F})ln(\frac{T_F}{T_S})$$
$$\Delta S_S = 4m\frac{L_F}{T_F}ln(2)$$
$$\Delta S_S = \frac{28}{10}m\frac{L_F}{T_F}$$

For the liquid:

$$\Delta S_L = -\frac{\frac{3}{4}mL_F}{T_F} + \int_{T_F}^{T_L} mC_L\frac{dT}{T})$$
$$\Delta S_L = -\frac{3}{4}m\frac{L_F}{T_F} + m\frac{5L_F}{2T_F}ln(\frac{T_F}{T_L})$$
$$\Delta S_L = -m\frac{L_F}{T_F}(\frac{3}{4} - \frac{5}{2}ln(\frac{2}{3}))$$
$$\Delta S_L = -m\frac{L_F}{T_F}(\frac{3}{4} - \frac{5}{2}(\frac{7-11}{10}))$$
$$\Delta S_L = -m\frac{L_F}{T_F}(\frac{3}{4} + 1))$$
$$\Delta S_L = -\frac{7}{4}m\frac{L_F}{T_F}$$

The sum of both is :

$$\Delta S_{tot} = \Delta S_S + \Delta S_L$$
$$\Delta S_{tot} = (\frac{28}{10} - \frac{7}{4})m\frac{L_F}{T_F}$$
$$\Delta S_{tot} = \frac{21}{20}m\frac{L_F}{T_F}$$
$$\Delta S_{tot} = 1.05m\frac{L_F}{T_F}$$

The result $\Delta S_{total} > 0$ is **positive**. This is consistent with the **Second Law of Thermodynamics**, which states that the total entropy of an isolated system must increase during an irreversible process.

Problem 3

1.
   - **State A:** Defined by pressure P_A and volume V_A. The temperature is $T_A = \frac{P_A V_A}{nR}$.
   - **State B:** Since the process $A \to B$ is isothermal, $T_B = T_A$. Given $P_B = 2P_A$, we use Ideal Gas Law ($P_A V_A = P_B V_B$):
   $$V_B = \frac{P_A V_A}{2P_A} = \frac{V_A}{2} \tag{4}$$
   - **State C:** Since the process $C \to A$ is isovolumetric, $V_C = V_A$. Since the process $B \to C$ is isobaric, $P_C = P_B = 2P_A$. The temperature is determined by the ideal gas law:
   $$T_C = \frac{P_C V_C}{nR} = \frac{(2P_A)V_A}{nR} = 2T_A \tag{5}$$

2.
   - **Process $A \to B$:** Since the process is isothermal:
   $$W_{AB} = nRT_A \ln\left(\frac{V_B}{V_A}\right) = P_A V_A \ln\left(\frac{V_B}{V_A}\right) \tag{6}$$
   Substituting $V_B = \frac{1}{2}V_A$:
   $$W_{AB} = P_A V_A \ln\left(\frac{1}{2}\right) = -P_A V_A \ln(2) \tag{7}$$
   - **Process $B \to C$:** Since the process is isobar, at constant pressure $P_B = 2P_A$, then
   $$W_{BC} = P_B(V_C - V_B) = 2P_A\left(V_A - \frac{1}{2}V_A\right) = P_A V_A \tag{8}$$
   - **Process $C \to A$** Since the volume is constant ($dV = 0$):
   $$W_{CA} = 0 \tag{9}$$

   The total work done by the gas is:
   $$W_{net} = W_{AB} + W_{BC} + W_{CA} \tag{10}$$
   $$W_{net} = -P_A V_A \ln(2) + P_A V_A + 0 \tag{11}$$
   $$W_{net} = P_A V_A (1 - \ln(2)) \tag{12}$$

3. The cycle uses water vapor, a non-linear polyatomic ideal gas (6 degrees of freedom). Therefore, the molar heat capacities are:
   - $C_V = 3R$
   - $C_P = C_V + R = 4R$

   (a) **Process $A \to B$ (Isothermal):** $Q = nRT \ln\left(\frac{V_f}{V_0}\right)$
   $$Q_{AB} = P_A V_A \ln(1/2) = -P_A V_A \ln 2 \quad \text{(Heat released)} \tag{13}$$
   (b) **Process $B \to C$ (Isobaric):** $Q = nC_P \Delta T$
   $$Q_{BC} = n(4R)(T_C - T_B) = 4nR(2T_A - T_A) = 4P_A V_A \quad \text{(Heat absorbed)} \tag{14}$$
   (c) **Process $C \to A$ (Isochoric):** $Q = nC_V \Delta T$
   $$Q_{CA} = n(3R)(T_A - T_C) = 3nR(T_A - 2T_A) = -3P_A V_A \quad \text{(Heat released)} \tag{15}$$

- $Q_{in} = Q_{BC} = 4P_AV_A$
- $Q_{out} = |Q_{AB}| + |Q_{CA}| = P_AV_A(\ln 2 + 3)$

4. • Efficiency based on Net Work

$$\eta = \frac{W_{net}}{Q_{in}}$$

$$\eta = \frac{1 - ln(2)}{4} = \frac{1 - \frac{1}{2}}{4}$$

$$\eta = \frac{1}{8} = 0.125 = 12.5\%$$

   • Efficiency based on Heat Exchanged

$$\eta = 1 - \frac{Q_{out}}{Q_{in}}$$

$$\eta = 1 - \frac{ln(2) + 3}{4} = 1 - \frac{3.5}{4} = \frac{0.5}{4}$$

$$\eta = \frac{1}{8} = 0.125 = 12.5\%$$

5. To evaluate the performance of our heat engine, we compare its efficiency to that of a reversible Carnot engine operating between the same two extreme temperatures of the cycle.

   The extreme temperatures of the cycle are: $T_{hot} = T_C = 2T_A$ and $T_{cold} = T_A$

   The efficiency of a reversible Carnot engine is:

$$\eta_{Carnot} = 1 - \frac{T_{cold}}{T_{hot}} = 1 - \frac{T_A}{2T_A} = 0.5 = 50\%$$

   Therefore $\eta_{cycle} < \eta_{Carnot}$

   As expected, the efficiency of our heat engine is significantly lower than the Carnot efficiency. The Carnot engine represents the theoretical upper limit for any heat engine operating between two given temperatures.